

Grafting of glycidyl methacrylate onto polypropylene using supercritical carbon dioxide

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Abstract

Free-radical grafting of glycidyl methacrylate (GMA) onto polypropylene (PP) films has been studied using supercritical carbon dioxide (SC-CO₂) as a solvent and a swelling agent. As the reaction temperature was below the melting point, PP was modified in the solid phase. The PP film was first soaked with the monomer GMA and benzoyl peroxide (BPO) as an initiator using SC-CO₂ at different experimental conditions of pressure, temperature, and thermal treatment time. After releasing CO₂, film GMA molecules were grafted onto PP in different times. Using this method, the degree of grafting and the morphology could be controlled through the combination of pressure, temperature, and soaking time. FTIR spectra confirmed that GMA had been grafted onto PP and that polypropylene-graft-glycidyl methacrylate (PP-g-GMA) presented a high surface reactivity for conductive polyaniline anchoring. DSC measurements and TG analyses showed that the thermal profiles of the graft copolymer and virgin PP are quite similar and that the graft PP does not exhibit changes in terms of thermal degradation profile and melting temperature, respectively. X-ray data showed that a high degree of grafting leads to a lower degree of crystallinity of polypropylene.

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1. Introduction

The use of a supercritical fluid to soak and swell a polymer matrix with a reactant is a quite well-known procedure. For example, McCarthy's group published a number of papers concerning monomer grafting [1,2], the chemical modification of buried interfaces [3],

the synthesis of polymer/metal composites [4], and polymer/polymer composites [5], among others.

Polypropylene (PP) has been largely used in a number of industrial and technological fields, but its use is limited in some specific applications due to its low chemical versatility derived from the inexistence of reactive functional groups in its structure. In this sense, several efforts [6] have been made to increase surface-to-surface interactions for PP applications, including grafting monomers [7–11]. Glycidyl methacrylate (GMA) and benzoyl peroxide have been used as grafting initiators. According to the literature, [12,13] polymerisation

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occurs *via* C–C double bond cleavage and results in a graft material with the original reactivity of the epoxy ring. Thus, the epoxy group can be effectively used to anchor the desired species. Bernard and co-workers [14] used the epoxy reactivity to anchor cyclodextrins in graft-GMA polypropylene and studied their activity in wastewater treatment. Based on the reactivity of epoxy with amine and hydroxyl groups, [7,15] Huang and co-workers [13] developed a new route for grafting PP with GMA in order to obtain chemical compatibility in polymeric blending processes. Polypropylene, polyethylene, and polystyrene could also be grafted with GMA due to the interactions between the epoxy group and stabilizers such as 2,4-dihydroxybenzophenone, phenyl-4-aminosalicylate, and 4-amino-2,2,6,6-tetramethylpiperidine, which prevented the degradation of the polymer [7].

Supercritical fluid technology made extraordinary progress in the past decade in terms of commercial application and fundamental understanding of solution behaviour. An important contribution comes from the mounting environmental pressures on industry to replace volatile organic compounds [16] and ozone-depleting substances such as process solvents. This has led to “green” alternatives as recently reviewed [17,18]. Among the common advantages of using supercritical CO₂ medium (e.g., simplified reaction/separation schemes, lower energy requirements, and improved product quality), it is worth mentioning the ability of controlling the degree of grafting onto the polymeric matrix with a dry method or without an extrusion process [1].

As to the use of SC-CO₂, findings reported to date include the work by Liu and co-workers [19], who prepared isotactic PP grafted with methyl methacrylate (MMA) and observed a decrease in crystallinity as well as high grafting degree. Our recent efforts have dealt with the development of polymer mixtures using SC-CO₂ [20–23]. The present contribution concerns the study of the main parameters involved in grafting PP with glycidyl methacrylate and benzoyl peroxide as an initiator, and the remaining epoxy reactivity with amine groups.

2. Experimental

2.1. Materials

Thin PP films (50 μ m, Poliolefinas Co., Brazil) were cut to the dimension of 3 $\times$ 4 cm, extensively washed with acetone (Merck) for 12 h (Soxhlet), dried at 80 °C for 4 h, and kept under vacuum. The grafting process followed two individual steps: (1) soaking by glycidyl methacrylate (GMA, Aldrich) and the initiator (BPO, Riedel-de-Häen) by supercritical CO₂ and (2) thermal-induced grafting (a process substep).

2.2. Grafting procedure

One end of a tapped 1/16 in NPT thread vessel was plugged and the other was fitted with a 1/16 in high-pressure needle valve, giving the vessel a total volume of ca. 25 mL. The vessel was charged with a mixture of GMA (5.0 g) and BPO (0.026 g). Soon afterwards, the PP film (ca. 0.2 g) was added to the vessel (without physical contact with the GMA/BPO mixture) and the vessel air was pumped out. After the system had reached thermal equilibrium, CO₂ was pumped in (Thar Designs, Model P-50) until the desired pressure was reached. During this process, the PP films were soaked and infused with some of the GMA + BPO mixture. After the desired soaking time, the system fluid was purged. Some experimental parameters were determined by preliminary essays, Step 1: (a) soaking time of 4 h, 110 bar, and 50 °C, hereafter sample A, and (b) soaking time of 10 h, 130 bar, and 70 °C, hereafter sample B. The CO₂ pressure drop lasted 1 min. After soaking, the samples were washed with distilled water, dried with N₂ flux, and introduced in a second supercritical stainless steel vessel pressurized with N₂ in step 2 after ca. 30 min. All work in step 2 was done at 120 °C, 50 bar as follows. Samples A from step 1 were heated for: (i) 1 h, (ii) 2 h, (iii) 3 h, and (iv) 4 h. Based on sample A results (also cf. Ref. [19]), samples B from step 1 were heated for only 4 h. At the end of the reaction, the vessel nitrogen was purged and the samples were Soxhlet-extracted using acetone for 12 h to remove the unreacted monomer and homopolymer. The percentual degree of grafting was calculated by a procedure described elsewhere [19,24]. To investigate the epoxy reactivity of the graft PP [25,26], the samples were immersed into an aqueous acid solution containing aniline for 5 h at 1 °C [27–29].

2.3. Instrumentation

The grafting process was carried out using a Thar equipment, model SFE 500 and carbon dioxide (White Martins) with 99.95% purity. FTIR spectra were obtained using a spectrophotometer Bomem, model MB100. Differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA) were made in a Shimadzu apparatus, model 50 with N₂ flux at 20 mL min^{−1} and heating rate of 10 °C min^{−1}. X-ray diffraction experiments were carried out using a diffractometer Shimadzu, model D6000 with Cu K α radiation. The four-point probe technique [30] was used to measure surface conductivity. An atomic force microscope Shimadzu, model SPM-9500J3 was used to analyze surface morphology.

3. Results and discussion

The distortive effect of CO₂ soaking on the polymer matrix was not investigated in this work. GMA/BPO

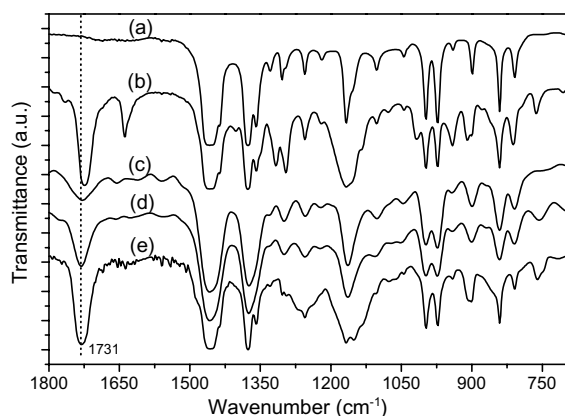


Fig. 1. FTIR spectra of (a) virgin PP, (b) GMA-soaked PP (step 1, sample A), (c) PP-g-GMA (step 2, sample A, 1 h thermal treatment), (d) PP-g-GMA (step 2, sample A, 4 h thermal treatment), and (e) PP-g-GMA (sample B, 4 h thermal treatment).

soaking (step 1) and grafting (step 2) were accompanied through FTIR spectroscopy (Fig. 1). Spectrum (a) corresponds to virgin PP and (b) to GMA/BPO-soaked PP. The latter presented noticeable peaks at 1723 and 1640 cm^{-1} , assigned to C=O and C=C, respectively, due to the GMA/BPO species [15,31–33]. GMA free radical grafting onto PP could be confirmed through the decreasing intensity of the peak at 1640 cm^{-1} (Fig. 1(c)) and the shift of the C=O signal to higher wavenumbers due to the loss of π conjugation in GMA (spectra (c), (d), and (e)). The spectra of thermally treated films (step 2) are shown in Fig. 1(c)–(e). Spectrum 1(e) is more resolved (optimal grafting conditions, cf. Ref. [19]) and shows broader signals in the 450–1500 cm^{-1} range attributed to a higher graft ratio of GMA to the PP-matrix since the characteristic PP and GMA signals are very close. The signals at 1148–1270 cm^{-1} were assigned to C–O–C stretching and in the 845–910 cm^{-1} range to epoxy stretching. The grafting percentages determined from weight variation for the samples corresponding to spectra (c), (d), and (e) are 1.1%, 2.9%, and 13.8%, respectively, which indicate progressive grafting as a function of heating time.

The efficiency of the grafting process was verified by exposing the resulting product to aminic compounds [7]. In this sense, the reactivity of the epoxy ring of polypropylene-graft-glycidyl methacrylate samples, hereafter PP-g-GMA, was studied by immersing them into a polymerising polyaniline (PANI) solution. After 5 h of reaction time, the films were extensively washed with *N*-methyl-2-pyrrolidone and dried at 80 °C for 4 h. The resulting films presented a greenish aspect characteristic of PANI emeraldine base [34] and were subsequently characterized by FTIR spectroscopy (Fig. 2). The longer the thermal treatment time was, the higher

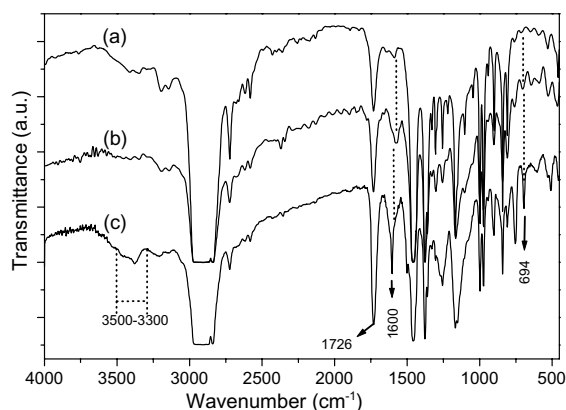


Fig. 2. FTIR spectra of PP-g-GMA films immersed in PANI polymerisation solution. (a) Sample A, 1 h thermal treatment, (b) sample A, 4 h thermal treatment, and (c) sample B, 4 h thermal treatment.

the intensity of the signal at ca. 1600 cm^{-1} was, which is characteristic of PANI [35] and not significant for the samples of step 1 and 2 (cf. Fig. 1). This feature evidences the deposition of a larger amount of PANI onto the graft samples submitted to longer thermal treatment times. This suggests an increasing degree of grafting, and consequently, an increasing number of reactive epoxy groups. Additional signals were observed for PANI in the 3300–3500 cm^{-1} region relative to the N–H stretching and at 694 cm^{-1} , relative to chlorine from HCl used as a doping agent [31].

DSC measurements were performed for virgin PP and for different samples (Fig. 3). It is clear from the melting behaviour that the PP-g-GMA copolymer presents a double melting feature, the first being broader and lower than the second one. According to data in the literature [19], this indicates that the copolymer obtained is composed of PP-g-GMA and PP components. Based on reported data [36], the first melting process (ca. at 120 °C) is assigned to molecular fractioning during temperature-induced crystallization; thus, it is not due to either the GMA or the BPO used in the formation of PP-g-GMA, but to the thermal treatment performed during the grafting process. In addition, no significant differences were observed in T_m (at 161 °C) for PP-g-GMA samples compared to virgin PP. Otherwise, considering the FWHM (full-width at half-maximum height) of the second T_m signal, one can observe a lower percentage of FWHM for graft samples (b) and (c) compared to virgin PP (curve (a)); however, the FWHM of curve (d) is comparable to that of the virgin PP curve. These results suggest that samples A heated for 4 h present a higher degree of crystallinity than virgin PP does. However, the broadening fusion peak at 161 °C for curve (d) indicates an increasing uniformness of the lamellar distribution [19] compared to virgin PP. These

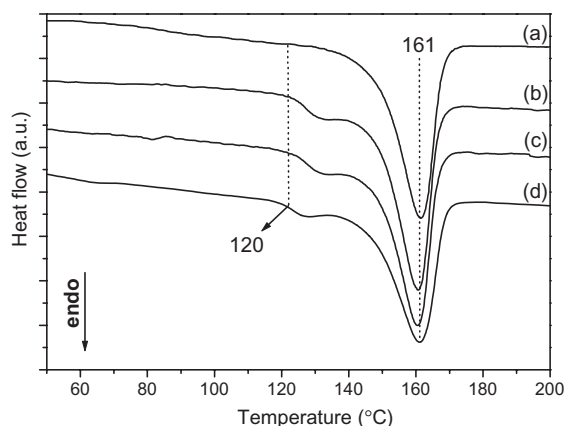


Fig. 3. DSC curves of (a) virgin PP, (b) sample A, 1 h thermal treatment, (c) sample A, 4 h thermal treatment, and (d) sample B, 4 h thermal treatment.

results agree with data in the literature for PP-g-MMA (polypropylene-graft-methylmethacrylate) [19].

The thermal stability is illustrated in the weight loss curves (Fig. 4). The weight loss of the graft samples began at 200 °C, i.e., before that of virgin PP. According to the literature [37], this feature can be assigned to the loss of poly(GMA) grafted onto the PP surface. Except for the beginning of the weight loss curves, the weight loss temperature of the graft copolymer corresponding to curves (b) and (c) (samples A) are higher than that of virgin PP. Moreover, considering the sample corresponding to curve (c) as the one with the highest GMA concentration as suggested by FTIR results, the similarities between curve (c) and the weight loss curve of poly(GMA) reported elsewhere [37] can be clearly understood. These features indicate the high thermal stability of PP-g-GMA corresponding to sample A from

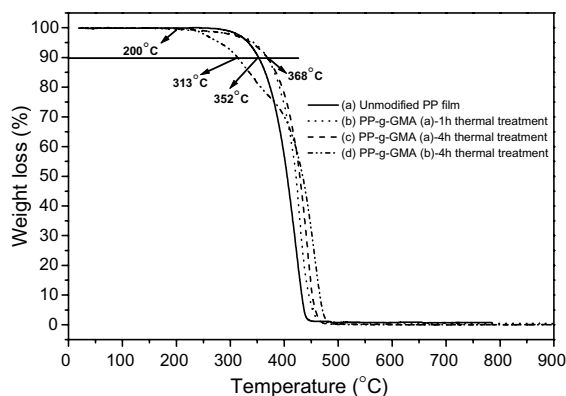


Fig. 4. TG thermographs of (a) virgin PP, (b) sample A, 1 h thermal treatment, (c) sample A, 4 h thermal treatment, and (d) sample B, 4 h thermal treatment.

step 1 (4 h soaking, 110 bar, 50 °C) is unlike that of sample B (10 h soaking, 130 bar, 70 °C) (curve (d)). This fact is credited to the high GMA content of sample B, which led to a thermal degradation profile similar to that of GMA [38].

The X-ray diffraction pattern of virgin PP was compared to the diffractogram of the sample prepared under maximum grafting conditions (sample B, 4 h thermal treatment), Fig. 5(a) and (c), respectively. These diffraction patterns suggest that the grafting process alters the crystallinity of PP; nevertheless, this alteration is not assigned to the grafting process in particular, but to the thermal treatment. This hypothesis can be confirmed through diffractogram 5(b) of polypropylene submitted to similar conditions, except for the introduction of GMA and BPO into the reactional medium, hereafter reference-PP sample, rPP. Fig. 5(b) clearly presents well-defined peaks, indicating a crystallinity higher than that of virgin PP. The data for degree of crystallinity calculated from deconvoluted curves are shown in Table 1. The two diffraction peaks appearing at $2\theta = 14^\circ$ and $2\theta = 21^\circ$ indicate a β crystal form. After the thermal treatment, rPP presented a lower degree of crystallinity than that of virgin PP, correlating to β -phase and indicating the formation of a uniform α crystal form [19].

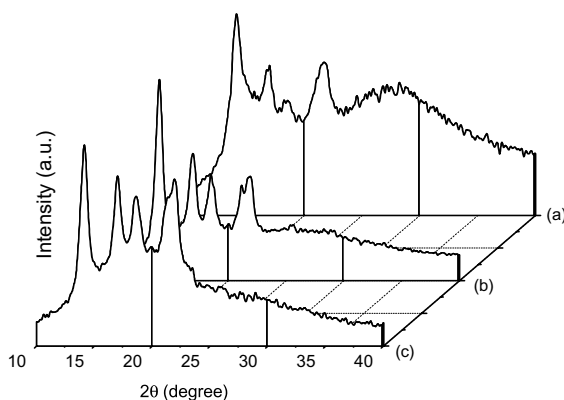


Fig. 5. X-ray diffraction patterns of (a) virgin PP, (b) reference-PP (similar conditions as those of sample B, without GMA/BPO), and (c) PP-g-GMA (sample B, 4 h thermal treatment).

Table 1

Crystallization behaviour data of virgin PP, reference-PP, and PP-g-GMA (sample B, 4 h thermal treatment)

Sample	Integrated area under 2θ signal (arbitrary unit)	
	$2\theta = 14^\circ$	$2\theta = 21^\circ$
Virgin PP	1.8	0.4
rPP	0.9	0.2
PP-g-GMA	1.0	0.3

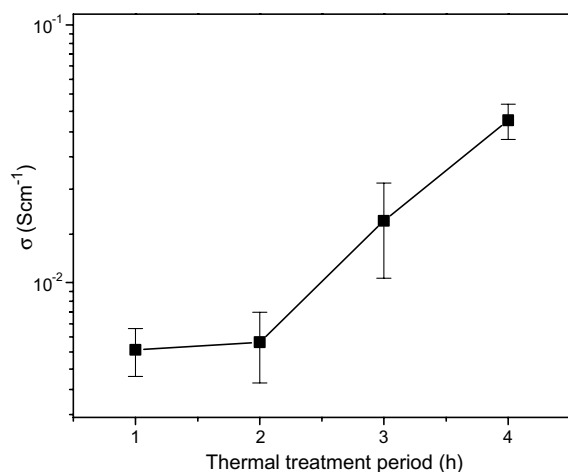


Fig. 6. Surface conductivity of PP-g-GMA films immersed in PANI polymerisation solution as a function of thermal treatment time during grafting.

These results also suggest that PP-g-GMA undergoes an increase in the amount of β crystal form when compared to rPP, which agrees with the literature [39] and DSC measurements (FWHM calculi). Moreover, according to the literature [39] and current X-ray data, an increasing degree of grafting leads to a lower degree of crystallinity.

Although the effect of peroxide on PP, mainly chain scissions, is known, the conclusion on crystallinity based on either DSC or thermal stability results does not change. Literature-based results [40] have shown that at low reaction temperatures, chain scission takes place to a much lesser extent as it is a monomolecular reaction and has a higher activation energy than other reactions do. In addition, Huang and Liu [13] have shown that termination by combination of macroradicals at low temperatures results preferably in the crosslinking of PP rather than chain scission. Moreover, we also suggest that the primary free radicals generated by the

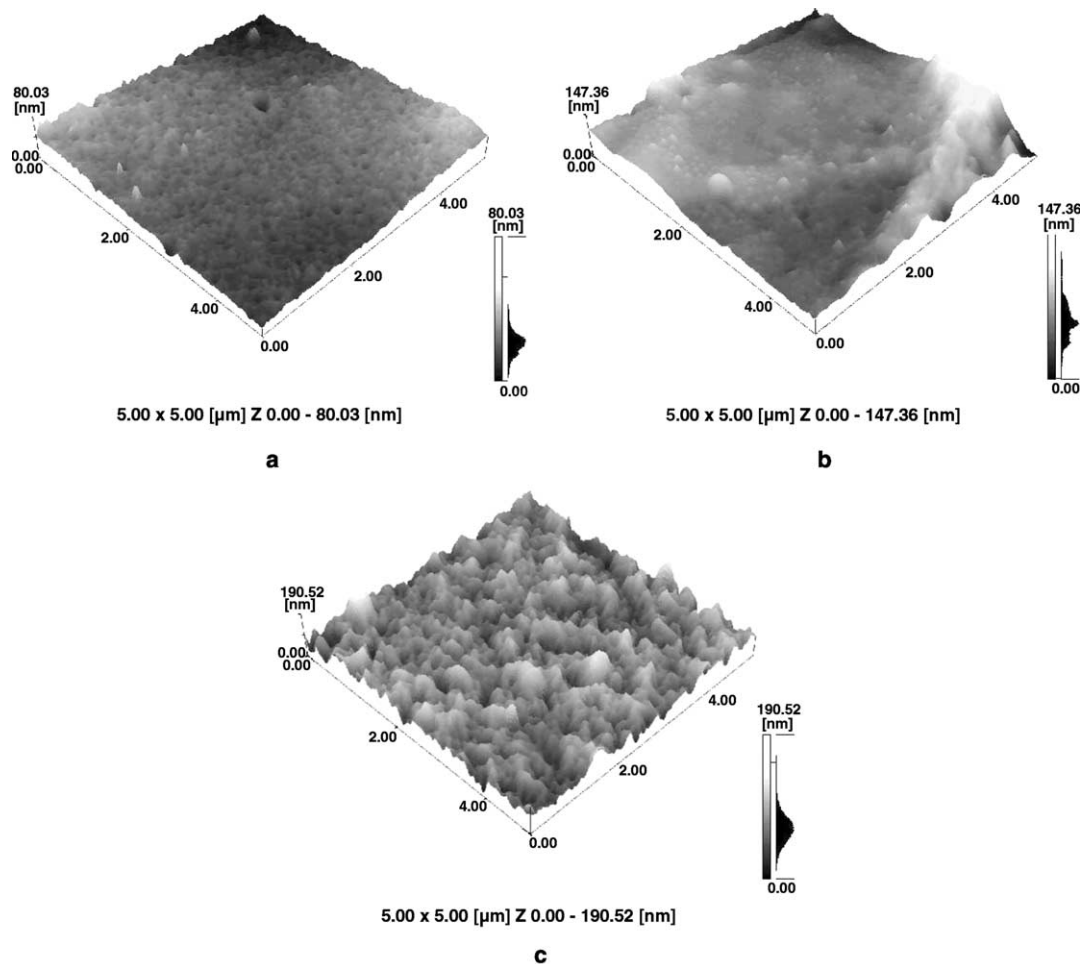


Fig. 7. AFM images of (a) virgin PP, (b) PP-g-GMA (sample B, 4 h thermal treatment), and (c) PP-g-GMA (sample B, 4 h thermal treatment) immersed in PANI polymerisation solution.

decomposition of BPO have a higher tendency to attack GMA molecules than PP chains. Thus, a minor number of chain scissions would not affect the previous statements.

Fig. 6 illustrates the surface conductivity (σ) of samples A (1, 2, 3, and 4 h thermal treatment). As expected, the longer the thermal treatment, the higher the value of σ . PP-g-GMA/PANI prepared using sample B (10 h soaking, 4 h thermal treatment) presented the highest surface conductivity, ca. $1.3 \pm 0.07 \text{ S cm}^{-1}$, not shown in Fig. 6. Both aforementioned facts are assigned to progressively higher GMA species incorporated onto PP as suggested by FTIR results. The conductivity results of protonated PANI [27,34] ($1\text{--}10 \text{ S cm}^{-1}$) agree with data in the literature.

Film surface was characterized by AFM analysis (Fig. 7). Compared to PP-g-GMA, virgin PP presented a smoother surface and a more continuous roughness distribution. Moreover, PANI adhesion had a relevant effect on the morphology and homogeneity of the film surface, which presented a rougher profile, probably attributable to the conductive polymer (Fig. 7(c)).

4. Conclusions

The PP-g-GMA graft copolymer was prepared with a free radical polymer in swollen PP matrix with the aid of supercritical CO_2 (SC- CO_2) as a solvent of GMA/BPO and as a swelling agent of the PP matrix. A valuable advantage in using SC- CO_2 medium for grafting GMA onto PP was the possibility of controlling the degree of grafting by simply changing experimental conditions. This feature avoided the use of an extrusion process or a wet chemical method.

FTIR spectra confirmed that GMA had actually been grafted onto PP. In addition, PP-g-GMA presented a high surface reactivity to PANI anchoring. Unlike in a previous report on PP-g-MMA, DSC measurements showed that the melting temperatures of the graft copolymer and virgin PP were quite similar. The graft PP did not exhibit changes in thermal degradation profile. Moreover, DSC results agree with X-ray data very well, showing that a subtle thermal treatment was enough to render PP more crystalline. However, high graft levels led to a lower degree of crystallinity of polypropylene. TG analyses showed that the higher the GMA content, the more similar the thermal behaviour of pure GMA and PP-g-GMA were. A progressive surface conductivity value was assigned to a higher concentration of epoxy-anchorage species on PP. AFM studies allowed to assess the effect of grafting on the surface morphology. The present study has shown that PP-g-GMA has potential applications in electromagnetic interference (emi) shielding and/or anti-static electricity coatings.

Acknowledgements

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